

MANOJRAJA BAKKIYARAJA

R&D Full Stack Engineer – AI Integrations & Automation

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SUMMARY

R&D-focused AI Full Stack Engineer with around 5 years of experience in building scalable web applications, automation, and AI integrated solutions. Have a proven track record of transforming one-line ideas into production-ready solutions by rapidly developing proofs of concept (PoCs) and minimum viable products (MVPs)—and scaling them into complete solutions. Expertise includes building Agentic AI (GenAI)-powered tools, modernizing legacy systems, and automating complex workflows. I am proficient in React, Flask, and MongoDB, delivering scalable, high performance applications from design to deployment. Recognized for operating independently with full ownership across design, development, documentation, delivery and maintainability—consistently ensuring measurable impact.

EXPERIENCE

Technology Analyst

Infosys Ltd

07/2024 - Present | Bangalore (Remote)

Remote role focusing on advanced automation, web development, AI integrations and Mentorship

- Developed and deployed a full-stack Wi-Fi testing automation platform, replacing legacy tools, achieving 90% manual effort reduction, with over \$0.5 million quarterly savings through comprehensive ownership.
- Built a GenAI RAG web app using LangChain and ChromaDB to automate user story, acceptance criteria & test case generation, cutting manual effort by 80%.
- Engineered a GenAI Streamlit application that analyzes massive Excel datasets (100K+ rows, 200+ columns), dynamically grouping data, generating automated insights like top-N scenarios and KPI dashboards. This innovation slashed analysis time from days/weeks to minutes.
- Converted flat Python scripts to modular Flask APIs with real-time test execution.
- Mentored juniors in backend and GenAI development, fostering technical growth and hands-on learning.

Senior Systems Engineer

Infosys Ltd

04/2023 - 06/2024 | Bangalore (Remote)

Remote role focusing on automation, code migration and AI integrations

- Built AI-powered capabilities into an application, automating tasks such as: providing a natural language data assistant for MongoDB, generating API code, and summarizing reports. This enhances data accessibility and productivity.
- Built a GenAI chatbot for dynamic code generation in Flask, Pytest, and Robot Framework.
- Developed a RAG-based chatbot using Gemini API & multiple local LLM Models with vector search to query uploaded documents and images.
- Developed an NLP-based app to clean and condense large datasets, reducing 30,000+ rows down to 200 to 500 rows based on configurable thresholds for faster, focused analysis.
- Migrated backend from Express.js to Flask and mentored a junior developer to further upgrade the Flask codebase to FastAPI for improved performance and maintainability.
- Automated Wi-Fi log comparison by enabling bulk upload of multiple log folders and generating consolidated Excel reports with charts and dashboards — cutting 2-3 weeks of manual effort down to just minutes.

Systems Engineer

Infosys Ltd

09/2021 - 03/2023 | Bangalore (Remote)

Remote role focusing on continuous learning, automation & web development

- Automated WiFi log parsing and Excel report generation, achieving a 98% reduction in manual time.
- Resulted in a reduction of manual processing time from 3 days to 2 minutes.
- Integrated Wi-Fi performance testing features into the application by reusing existing APIs, adding database support, and building interactive UI dashboards.

System Engineer Trainee

Infosys Ltd

05/2021 - 09/2021 | Mysore (Remote)

Trainee role focusing on full stack training.

- Trained in Java, Spring Boot, React, and SQL databases.
- Led the team & delivered a full-stack capstone project with a score 80/80.
- Maintained a technical evaluation score of 84% and was recognized as a top performer.

SKILLS

React.js	HTML	CSS	MUI	JavaScript
REST API	Flask	Python	Pandas	LLM
MongoDB	NoSQL	LangChain	AIML	
RASA	Prompt Engineering	Generative AI		

KEY ACHIEVEMENTS

Performance Excellence & Recognition

Rated 'Exceeded Expectations' for 3 consecutive years (2022–2024) for consistent high performance and delivery excellence.

Quantifiable Client Impact & Efficiency Gains

Independently delivered 10+ projects, including one that saved over \$0.5 million every quarter.

Achieved more than 90% reduction in manual effort across projects, boosting operational efficiency.

Awards & Innovation Leadership

- Received INSTA Award at Infosys for outstanding contribution to automation framework development.
- Earned Be the Navigator Business Incubator Certification at Infosys for submitting an innovative business idea, where I led and mentored a colleague on a GenAI-based Question Paper Generation (PoC).

Professional Development & Skill Expansion

Successfully completed 10+ courses and certifications in full-stack development and AI integration over the last 3 years, continuously expanding technical expertise.

PERSONAL PROJECT

Billing & Inventory Management System

08/2024 - 02/2025

Built a full-stack, GST-compliant POS system focused on billing, inventory, and reporting using modern technologies.

- Developed features including sales and purchase invoicing, stock tracking, tax calculations, credit & repayment tracking, and dashboard analytics.
- Integrated a natural language chatbot for real-time sales and stock queries, enhancing usability and accessibility.
- The system reduced manual effort by over 70%, minimized errors by 90%, and cut operational costs by up to 40% by automating reporting and eliminating repetitive Excel-based processes.

EDUCATION

Bachelor of Engineering (B.E.) - ECE	CGPA
Rajalakshmi Engineering College	7.8 / 10
08/2016 - 05/2020	Chennai

Higher Secondary Certificate (HSC) - Biology Stream	%
SKV Educational Institutions	91.2 / 100
04/2015 - 03/2016	Namakkal

Secondary School Leaving Certificate (SSLC)	%
SKV Educational Institutions	96.8 / 100
04/2013 - 03/2014	Namakkal